

Computer Science 40S-007

Rummy Card Game

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1. Introduction and Motivation

For the final project, we have decided to create a card game called rummy. We picked this particular card game because it is not too simple or complex; it suits our programming level. It is also a perfect game to play against a computer, allowing for the two of us to code different sections of the program.

2. Background and Related Work

a. Related Work: Rummy, the card game

Rummy is a card game that can be played between two or more people. The object of the game is to 'meld' your cards into two types of combinations.

Runs: These are consecutive sequences of three or more cards of the same suit.

Sets: These are collections of three or more cards of the same rank (but differing suits)

For example, a run can consist of one six of hearts, one seven of spades, and one eight of clubs. Similarly, a set may consist of three sixes of different suits (examples: hearts, spades, and clubs).

You can also lay off cards in rummy. Laying off is when a player puts down a card that matches the other player's meld. For example, if player 1 puts down a set of three eights, player 2 can put down another eight to add to the meld. This is another way for a player to get rid of their cards.

At the start of the game, the cards are shuffled by the dealer and each player is dealt ten cards. The remaining cards are placed in the center of the table to form the stock. The first card of the stock is placed faced up beside the pile to start off the discard pile (this leaves us with two different piles: the stock and the discard pile).

The user will go first, drawing a card from either the stock or the discard pile. Then the player will see if they can create any melded combinations or lay off one (or more) of their cards. If the

player cannot make any combinations or lay off a card, then they must discard one of their cards. If the player can meld some of their cards, they will create as many melds as possible and/or lay off a card; after this, the player will discard one of their cards, and it's the next player's turn, following the same rules.

The first person to sort their entire hand into the above combinations with a final card to place in the discard pile wins the game.

3. Problem Statement

Our goal is to simulate a rummy game between the player and the computer.

4. Proposed Solution Strategy and Implementation

We plan to complete our online Rummy card game by dividing the development tasks effectively between the two programmers, Ruhani and Celeste. The game will follow the rules outlined in the Background section, with both the player and computer taking turns to play. We plan to implement our ideas one step at a time. Here are some of our key components:

a. Card deck and shuffling

- We will create a standard 52 card deck. Each card will have a rank out of the thirteen possible ranks (A, 2, 3, ..., 10, J, Q, K) and one of four suits (Spades, Clubs, Diamonds, Hearts).
- We will create a shuffle function as well which will be called in the beginning of each game to ensure fairness.

b. Program Computing Logic

- The game will be designed for two players: one user/human player, and one computer player
- The “dealer” program will include drawing cards, creating valid melds (sets or runs), laying off cards, and discarding a card at the end of each turn.
- A round will end if a player wins through successfully melding all their cards, or when a player has placed a card in the discard pile.

c. Meld Detection

- The program will include a function to detect valid melds in the player and computer's hand.
- This will be used to validate the player's moves and there will be an error message telling the player to retry their turn if their input is unsuccessful.

d. Instructions

- In the beginning of the game, there will be an introductory message which will have instructions.
- Once the player is ready to play, there will also be instructions on how to input valid melds and how to show your turn so the program detects them.
- Also, the player can type in “instructions” at any time to see the set of instructions again.

e. User Interface

- After every turn, the user will see an updated version of their hand and the stock/discard pile.
- Depending on how things turn out, this may be a simple print to the console or possibly even animated.

f. Player vs. Computer Logic:

- Player (User): The player will be able to choose cards to draw and discard, and the system will check if the cards form valid combinations.
- Computer (AI/Algorithms): The computer will follow a simplified decision-making algorithm to draw from the discard pile or stock (maybe randomized if we don’t have time to code it), create melds, and discard a card based on available choices.

5. Resource Requirements

a. Development Tools:

- Programming Language: Python
- Integrated Development Environment (IDE): We can use Visual Studio Code, PyCharm, Thonny, or any text editor that supports Python.
- Version Control: GitHub will be used for version control, allowing us to collaborate effectively and track changes made by each member.

b. Hardware:

- A computer/laptop for each team member with Python installed.
- Access to the internet for research and collaboration.

c. Additional Learning Resources

- We may need something like YouTube or some other learning resources as our current knowledge may not be sufficient for this project.

6. Timeline of Activities

Activity	Programmer	End Time
Develop card deck and shuffling function	Celeste	February 4
Design and implement the instructions screen (game introduction)	Ruhani	February 4
Create and design game interface (player's view for each turn)	Ruhani	February 13
Integrate UI with game logic (meld detection, laying off cards, etc.)	Both	February 18
Implement meld detection logic (valid sets and runs)	Celeste	February 25
Implement card laying logic (allow players to add to existing melds)	Celeste	February 25
Win Detection logic (identify winning hands)	Ruhani	March 4
Design and implement end-of-game screen (show winner and stats)	Ruhani	March 11
User testing and debugging (playtesting to ensure smooth gameplay and fix issues)	Both	March 18
Final polish of game design, UI, and UX improvements	Both	March 25
Game launch and presentation preparation	Both	March 25

7. Expected Outcome

By the end of the project, we expect to have a fully functional Rummy card game where:

- **The Player Can Play Against the Computer:** The user should be able to play a full game against the computer, making valid moves like drawing, melding, and discarding cards.
- **Computer AI Performs Logical Actions:** The computer will be able to make decisions based on available cards and apply simple strategies to play the game (though probably not at an expert level).
- **Valid Gameplay and Win Condition:** The game will check for valid melds and ensure that the game ends when one player has completed all their melds and discarded their last card.
- **User Interface:** A simple CLI that is intuitive, displaying game progress and prompting for player actions. By implementing this project, we aim to demonstrate our understanding of the learning done in our CS40S class.